16S rRNA Amplicon Sequencing Analysis using QIIME 2

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Aim

To process and analyse 16S rRNA gene amplicon sequencing data using the QIIME 2 pipeline: to denoise the raw reads into amplicon sequence variants (ASVs), assign taxonomy, compare the alpha and beta diversity of microbial communities from four body sites (gut, left palm, right palm and tongue), and identify the bacterial taxa that differ significantly in abundance between two subjects in the gut samples.

Overall Workflow

1. Data preparation: import the multiplexed single-end reads and barcodes into QIIME 2 and demultiplex them into per-sample sequences.
2. Denoising: quality filtering, denoising and chimera removal with DADA2 to obtain the ASV feature table and representative sequences.
3. Feature table and phylogeny: summarise the feature table and representative sequences, then build a rooted phylogenetic tree (MAFFT alignment and FastTree).
4. Taxonomic analysis: classify the ASVs against the SILVA 138 (99%) naive Bayes classifier and draw taxonomic barplots.
5. Diversity analysis: alpha rarefaction, core phylogenetic diversity metrics, alpha group significance (Faith's PD, Shannon; Kruskal-Wallis), and beta group significance (Bray-Curtis; PERMANOVA) with PCoA ordination.
6. Differential abundance: ANCOM-BC on the gut samples, comparing subject-2 with subject-1 at ASV and genus level.

5.(A) Data Preparation and Pre-processing

Step 1: Create the directory and copy the files

```
mkdir emp-single-end-sequences
cp sequences.fastq.gz barcodes.fastq.gz emp-single-end-sequences/
ls
```

Step 2: Import the sequences

```
qiime tools import --type EMPSingleEndSequences --input-path emp-single-end-sequences --o
utput-path emp-single-end-sequences.qza
```

Step 3: Demultiplex the reads

```
qiime demux emp-single --i-seqs emp-single-end-sequences.qza --m-barcodes-file sample-met
adata.tsv --m-barcodes-column barcode-sequence --o-per-sample-sequences demux.qza --o-err
or-correction-details demux-details.qza
```

Step 4: Summarize and export demultiplexed sequences

```
qiime demux summarize --i-data demux.qza --o-visualization demux.qzv
qiime tools view demux.qzv

qiime tools export --input-path demux.qza --output-path demux_export
ls demux_export
```

Step 5: DADA2 - Denoising and ASV generation

```
qiime dada2 denoise-single --i-demultiplexed-seqs demux.qza --p-trim-left 0 --p-trunc-len
120 --o-representative-sequences rep-seqs-dada2.qza --o-table table-dada2.qza --o-denoisi
ng-stats stats-dada2.qza

qiime metadata tabulate --m-input-file stats-dada2.qza --o-visualization stats-dada2.qzv
qiime tools view stats-dada2.qzv
```

5.(B) Feature Table and Phylogenetic Analysis

Step 1: Summarize the ASV table and representative sequences

```
qiime feature-table summarize --i-table table-dada2.qza --o-visualization table.qzv
qiime feature-table tabulate-seqs --i-data rep-seqs-dada2.qza --o-visualization rep-seqs.
qzv
qiime tools view table.qzv
qiime tools view rep-seqs.qzv
```

Step 2: Generate the phylogenetic tree

```
qiime phylogeny align-to-tree-mafft-fasttree --i-sequences rep-seqs-dada2.qza --o-alignme
nt aligned-rep-seqs.qza --o-masked-alignment masked-aligned-rep-seqs.qza --o-tree unroote
d-tree.qza --o-rooted-tree rooted-tree.qza
```

5.(C) Taxonomic Analysis

Step 1: Assign taxonomy using SILVA

The full-length SILVA 138 99% classifier (compatible with QIIME 2 2024.10, scikit-learn 1.4.2) was used.

```
mkdir -p ~/Downloads/SILVA2026
cd ~/Downloads/SILVA2026
wget https://data.qiime2.org/classifiers/sklearn-1.4.2/silva/silva-138-99-nb-classifier.q
za
cd ~/bio-env
qiime feature-classifier classify-sklearn --i-classifier ~/Downloads/SILVA2026/silva-138-
99-nb-classifier.qza --i-reads rep-seqs-dada2.qza --o-classification taxonomy.qza --p-rea
ds-per-batch 2000
```

Step 2: View the taxonomy

```
qiime metadata tabulate --m-input-file taxonomy.qza --o-visualization taxonomy.qzv
qiime tools view taxonomy.qzv
```

Step 3: Taxa barplot

```
qiime taxa barplot --i-table table-dada2.qza --i-taxonomy taxonomy.qza --m-metadata-file
sample-metadata.tsv --o-visualization taxa-bar-plots.qzv
qiime tools view taxa-bar-plots.qzv
```

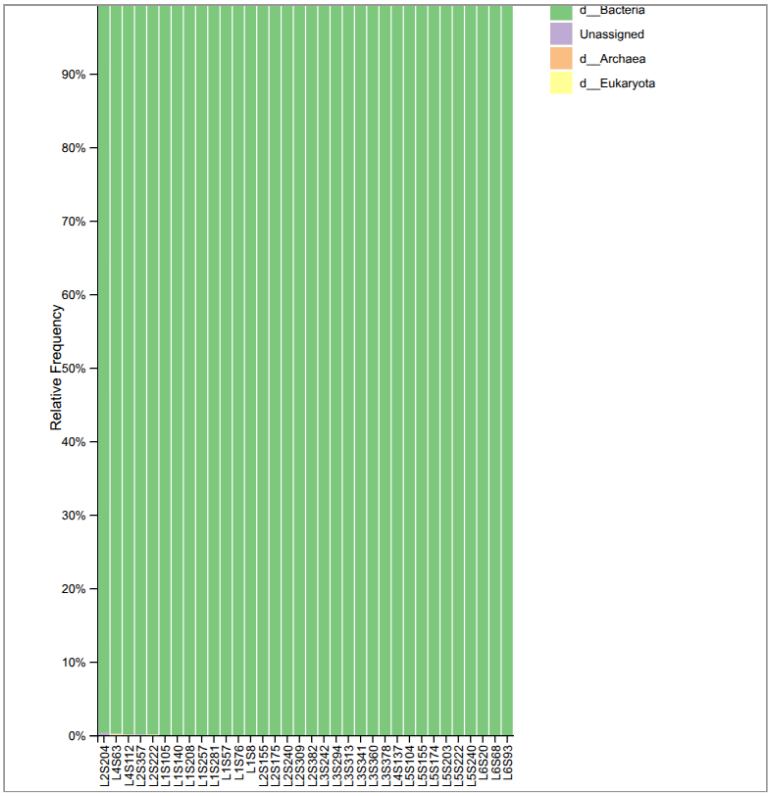


Figure 1. Taxa barplot at Level 1 (Domain), relative frequency per sample.

Interpretation: At the domain level every sample is almost entirely green, i.e. about 100% of the reads are classified as *Bacteria*. The legend also lists Unassigned, Archaea and Eukaryota, but these are visible only as negligible slivers at the base of a few bars. This shows that the 16S primers amplified bacterial DNA specifically, with little non-target amplification and almost no reads left unclassified, so the dataset is suitable for the bacterial diversity analysis that follows. A single domain gives no information on community differences, so finer taxonomic levels are needed for that.

5.(D) Diversity Analysis

Alpha rarefaction

```
qiime diversity alpha-rarefaction --i-table table-dada2.qza --i-phylogeny rooted-tree.qza
--p-max-depth 9820 --m-metadata-file sample-metadata.tsv --o-visualization alpha-rarefact
ion.qzv
qiime tools view alpha-rarefaction.qzv
```

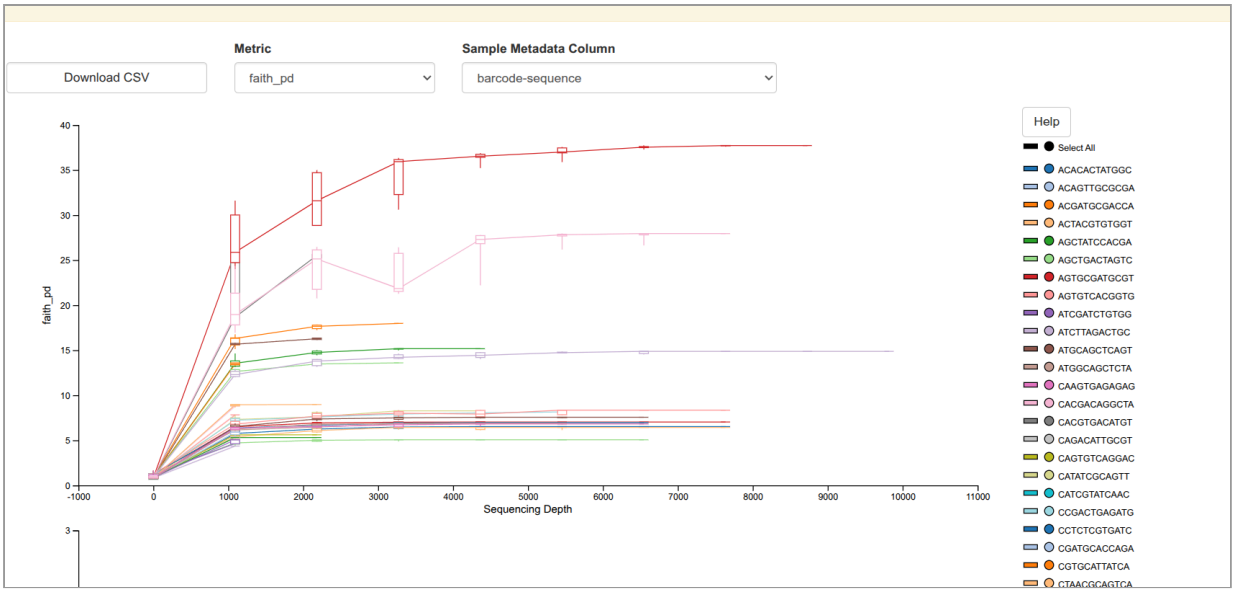


Figure 2. Alpha rarefaction curves (metric: faith_pd; each line is one sample, labelled by its barcode sequence).

Interpretation: The curves show how Faith's phylogenetic diversity increases as more reads are sub-sampled from each sample. Diversity rises steeply up to about 1000 reads and then flattens, and most samples are at a plateau by roughly 2000-4000 reads, meaning that the diversity of these samples has been adequately captured. Most samples stay low (about 5-15), whereas two samples are far more diverse: one reaches about 38 and another about 28, and these still rise slightly at greater depth. The lines end at different depths because each sample was sequenced to a different depth. The small boxes at each depth are the variation across repeated random sub-samplings and are very narrow at higher depths, so the estimates are stable. The sampling depth of 897 chosen for the core metrics lies on the early part of the curves, but it keeps all 34 samples, and the sample ranking is expected to be maintained because most curves are already close to their plateau.

Step 1: Run core metrics

```
qiime diversity core-metrics-phylogenetic --i-phylogeny rooted-tree.qza --i-table table-d
ada2.qza --p-sampling-depth 897 --m-metadata-file sample-metadata.tsv --output-dir core-m
etrics-results
```

Alpha diversity significance

Faith's PD vs body site:

```
qiime diversity alpha-group-significance --i-alpha-diversity core-metrics-results/faith_p
d_vector.qza --m-metadata-file sample-metadata.tsv --o-visualization core-metrics-result
s/faith-pd-body-site.qzv
qiime tools view core-metrics-results/faith-pd-body-site.qzv
```

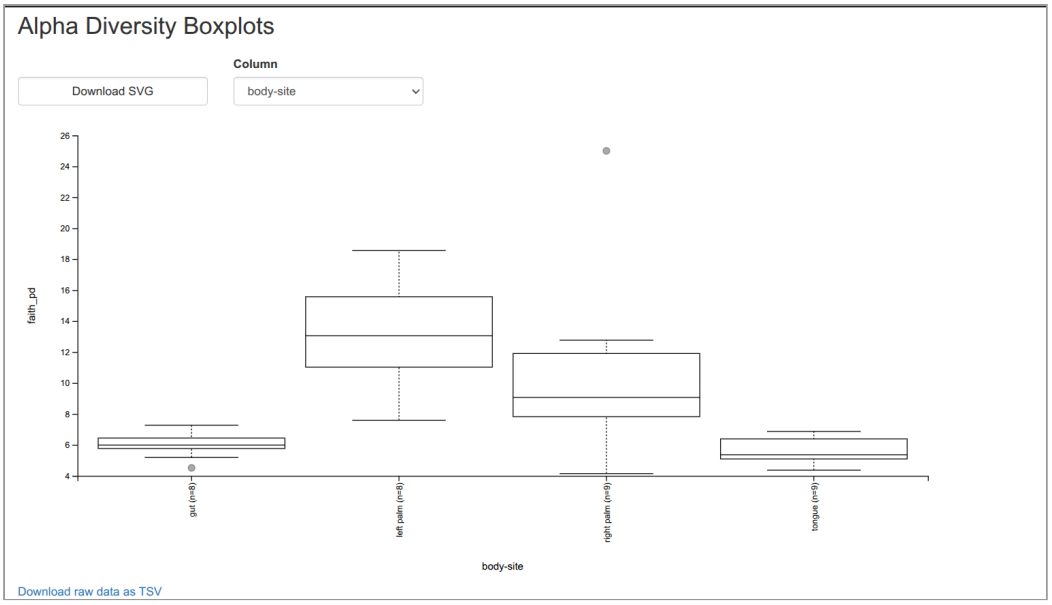


Figure 3. Faith's phylogenetic diversity (faith_pd) by body site.

Interpretation: Left palm has the highest phylogenetic diversity (median about 13, interquartile range about 11-15.5, maximum about 18.5). Right palm is intermediate and highly variable (median about 9, whiskers from about 4 to 13), with one outlier of about 25 that matches the highly diverse sample seen in the rarefaction plot. The gut (median about 6) and the tongue (median about 5.4, the lowest) have low diversity and very narrow boxes, so the individuals are similar to each other. Thus the two palm sites harbour phylogenetically broader and more individual-specific communities, whereas gut and tongue communities consist of a more restricted, closely related set of lineages. The boxes of left and right palm overlap, as do those of gut and tongue, while palm and tongue/gut differ clearly. With only 8-9 samples per group, the Kruskal-Wallis test run in this step is the formal test of these differences.

Shannon diversity vs body site:

```
qiime diversity alpha-group-significance --i-alpha-diversity core-metrics-results/shannon
_vector.qza --m-metadata-file sample-metadata.tsv --o-visualization core-metrics-results/
shannon-body-site.qzv
qiime tools view core-metrics-results/shannon-body-site.qzv
```

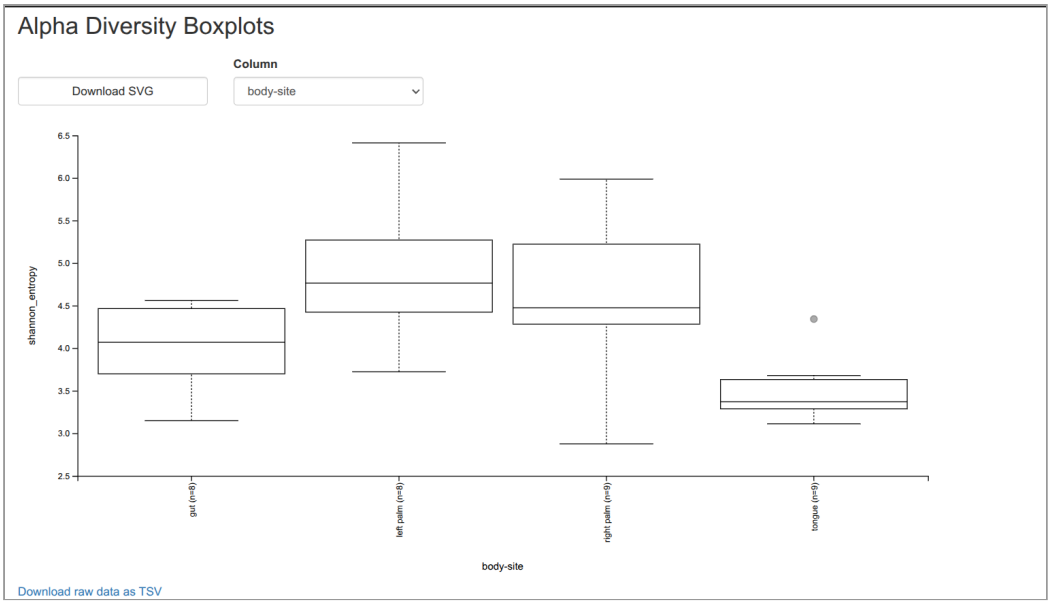


Figure 4. Shannon entropy (shannon_entropy) by body site.

Interpretation: Shannon entropy combines richness and evenness without using the phylogeny. Left palm again has the highest median (about 4.8), followed by right palm (about 4.5, with a wide range of about 2.9-6.0) and gut (about 4.1). The tongue is lowest (median about 3.4) with a narrow spread, indicating communities dominated by a few abundant taxa. The gut ranks relatively higher here than in Faith's PD: the gut contains many fairly evenly distributed ASVs, but they belong to closely related lineages, which gives it low phylogenetic diversity. The differences between sites are smaller and the boxes overlap more than for Faith's PD, so the evidence for differences between body sites is weaker for Shannon than for Faith's PD, except for the tongue, which is consistently lower.

Beta diversity significance

Bray-Curtis vs body site:

```
qiime diversity beta-group-significance --i-distance-matrix core-metrics-results/bray_cur
tis_distance_matrix.qza --m-metadata-file sample-metadata.tsv --m-metadata-column body-si
te --o-visualization core-metrics-results/bray-curtis-body-site.qzv --p-pairwise
qiime tools view core-metrics-results/bray-curtis-body-site.qzv
```

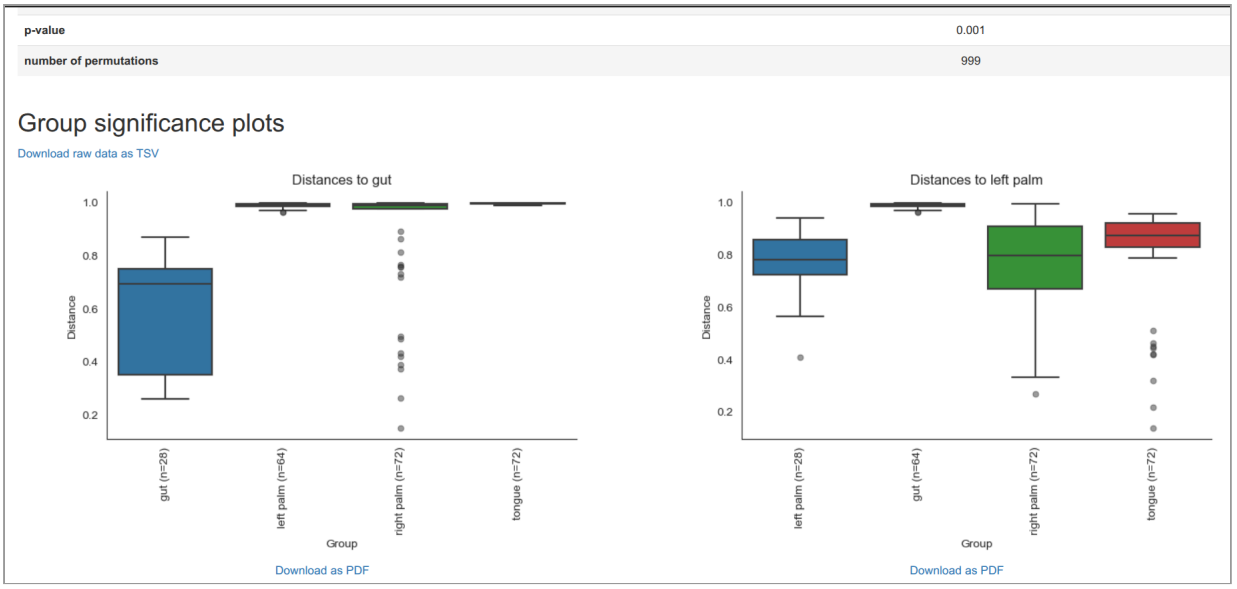


Figure 5. Bray-Curtis PERMANOVA result and group significance plots ("Distances to gut" and "Distances to left palm").

Interpretation: The PERMANOVA gives p = 0.001 with 999 permutations (the smallest p-value attainable with 999 permutations), so community composition differs significantly among body sites. In "Distances to gut", gut-to-gut distances are clearly smaller (median about 0.7) than the distances from gut to left palm, right palm and tongue, which are all close to 1.0, i.e. almost no shared ASVs between the gut and the other sites. In "Distances to left palm", left palm samples are closest to each other (median about 0.78) and to the right palm (median about 0.8, with a wide range), while they are about 1.0 from the gut and about 0.87 from the tongue. The two palms therefore resemble each other more than they resemble the gut or tongue, which is expected for similar skin habitats. Distances within a group are smaller than between groups, which is the pattern that a significant PERMANOVA reflects.

```
qiime tools view core-metrics-results/bray_curtis_emperor.qzv
```

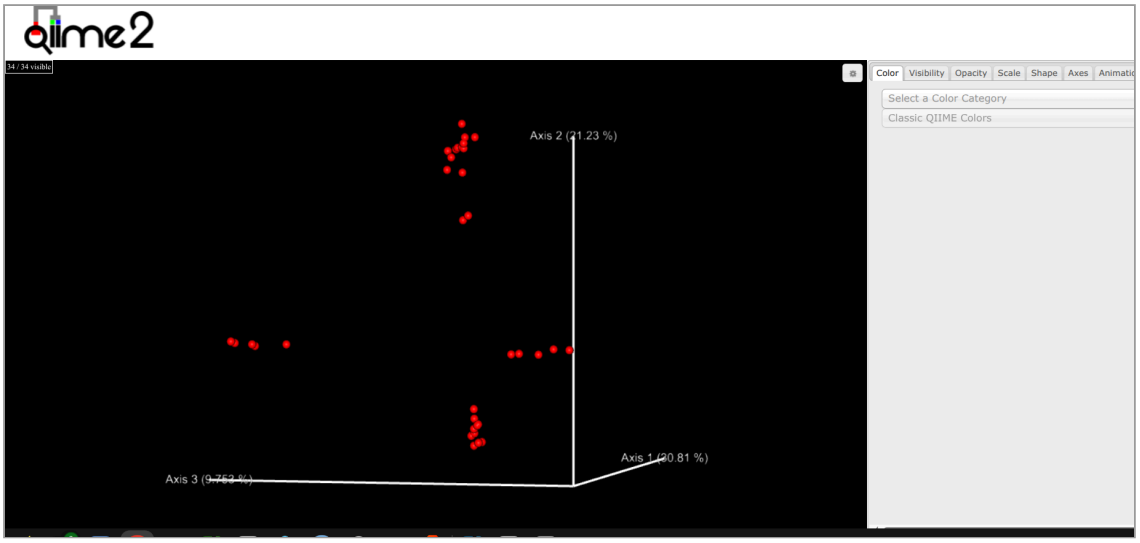


Figure 6. Bray-Curtis PCoA (Emperor), 34 samples, before colouring.

Interpretation: The first three axes explain 30.81%, 21.23% and 9.753% of the variation (about 61.8% in total), so the ordination represents the majority of the community differences. The 34

samples do not form one continuous cloud; they fall into several compact, well-separated clusters (upper, lower, left and centre-right in this view). Samples within a cluster have similar communities, and the clusters are far from each other, which is consistent with the large between-group Bray-Curtis distances seen in Figure 5.

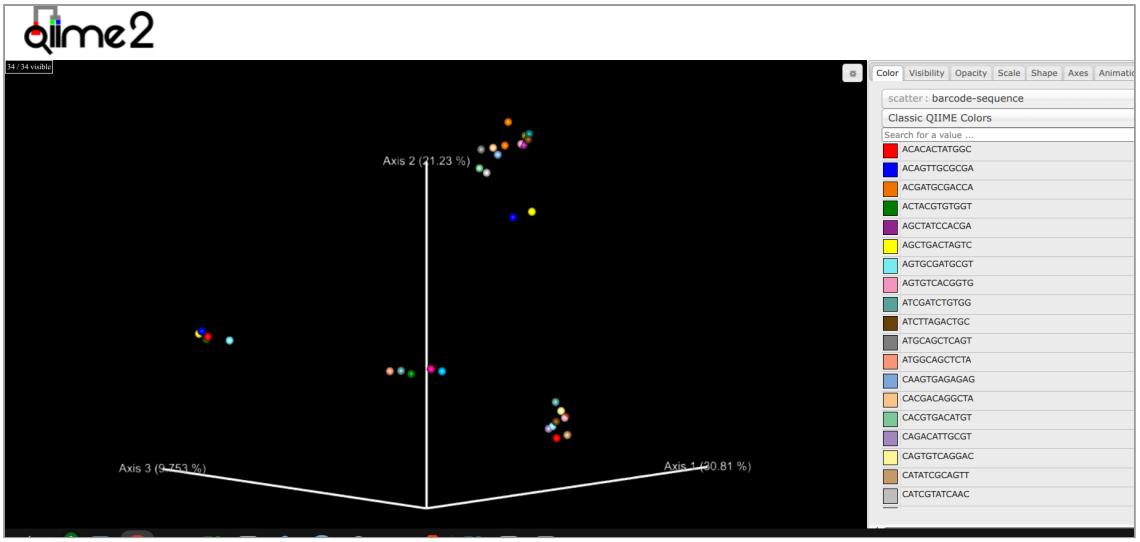


Figure 7. Bray-Curtis PCoA coloured by barcode-sequence.

Interpretation: Here each of the 34 points has its own colour because barcode-sequence is unique to every sample, so the colours identify individual samples and not body sites. The same cluster structure as in Figure 6 is visible, each cluster containing a mixture of colours. The grouping of samples into distinct clusters agrees with the significant PERMANOVA, and colouring by body-site would show which cluster corresponds to which site.

5.(E) Gut-Specific Differential Abundance Testing with ANCOM-BC

Step 1: Filter gut samples

```
qiime feature-table filter-samples --i-table table-dada2.qza --m-metadata-file sample-met
adata.tsv --p-where "[body-site]='gut'" --o-filtered-table gut-table.qza
```

Step 2: ANCOM-BC by subject (ASV level)

```
qiime composition ancombc --i-table gut-table.qza --m-metadata-file sample-metadata.tsv -
-p-formula "subject" --o-differentials ancombc-subject.qza
```

Step 3: Visualize the significant features (ASV level)

```
qiime composition da-barplot --i-data ancombc-subject.qza --o-visualization da-barplot-su
bject.qzv
qiime tools view da-barplot-subject.qzv
```

Step 4: Collapse gut table to genus (Level 7)

```
qiime taxa collapse --i-table gut-table.qza --i-taxonomy taxonomy.qza --p-level 7 --o-col
lapsed-table gut-table-l7.qza
```

Step 5: Run ANCOM-BC on genera

```
qiime composition ancombc --i-table gut-table-l7.qza --m-metadata-file sample-metadata.ts
v --p-formula "subject" --o-differentials l7-ancombc-subject.qza --verbose
```

Step 6: Make the genus-level plot

```
qiime composition da-barplot --i-data l7-ancombc-subject.qza --o-visualization l7-da-barp
lot-subject.qzv
qiime tools view l7-da-barplot-subject.qzv
```

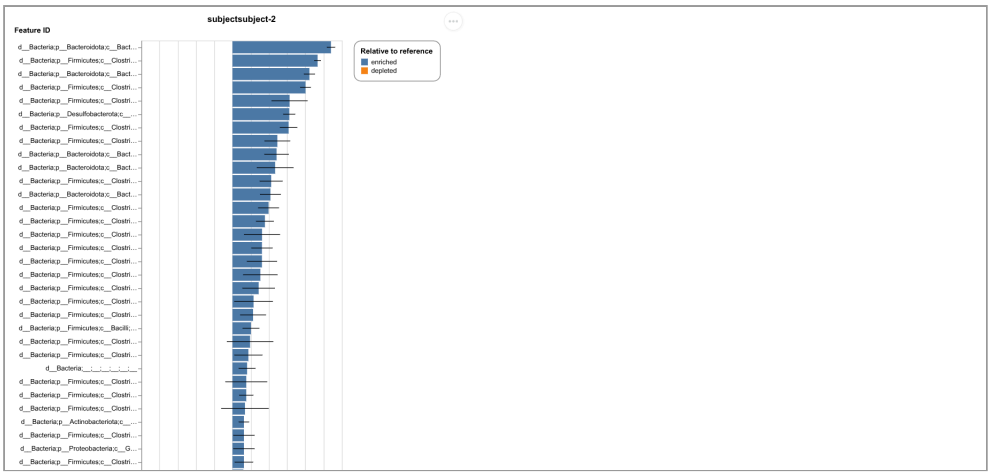


Figure 8. ANCOM-BC differential abundance barplot (subject-2 relative to subject-1), upper part of the plot.

Interpretation: The plot shows the log fold change (LFC) of each significant taxon in subject-2 relative to the reference subject-1, ordered from the largest LFC downwards, with the horizontal lines showing the standard error. All bars visible in this upper part are blue, meaning they are enriched in subject-2. Most of them are Firmicutes (class Clostridia) and Bacteroidota (class Bacteroidia), with single taxa from Desulfobacterota, Actinobacteriota and Proteobacteria and one taxon that is only classified as Bacteria. The largest effects are about 5 LFC units, and the bars become shorter with lower rank. Taxa with long error bars (shorter, lower bars) are less precisely estimated than the top taxa with short error bars. Because only two subjects are compared, these differences reflect inter-individual variation in the gut microbiota and not an effect of treatment.

Step 7: Export ANCOM-BC results

```
qiime tools export --input-path l7-ancombc-subject.qza --output-path ancombc-genus-export
```

Step 8: Find which genera are significantly different

```
python - <<'PY'
import csv
lfc_file = "ancombc-genus-export/lfc_slice.csv"
q_file = "ancombc-genus-export/q_val_slice.csv"
with open(lfc_file, newline="") as f:
    lfc = {row["id"]: float(row["subjectsubject-2"]) for row in csv.DictReader(f)}
with open(q_file, newline="") as f:
    q = {row["id"]: float(row["subjectsubject-2"]) for row in csv.DictReader(f)}
results = []
for feature in lfc:
    if q[feature] < 0.05:
        genus = feature.split(";")[-1].replace("g__", "")
        results.append((genus, lfc[feature], q[feature]))
results.sort(key=lambda x: x[2])
print("Significant genera (q < 0.05)")
print("-----------------------------")
print("Genus\tLFC\tq-value")
for genus, lfc_value, q_value in results:
    direction = "subject-2 enriched" if lfc_value > 0 else "subject-1 enriched"
    print(f"{genus}\t{lfc_value:.4f}\t{q_value:.4g}\t{direction}")
print("\nTotal significant genera:", len(results))
PY
```

Output: significant genera (q < 0.05)

Genus	LFC	q-value	Direction
s__	4.6715	2.316e-145	subject-2 enriched
s__	5.4050	1.112e-124	subject-2 enriched
s__	-4.2891	5.074e-56	subject-1 enriched
s__	4.2232	2.834e-43	subject-2 enriched
s__	4.0110	1.793e-41	subject-2 enriched
s__	3.1183	1.08e-19	subject-2 enriched
s__	3.0819	4.538e-09	subject-2 enriched
s__	-3.5267	0.009319	subject-1 enriched
s__	-1.9611	0.01363	subject-1 enriched
s__	1.7817	0.01772	subject-2 enriched
s__	2.0847	0.01917	subject-2 enriched
s__	2.4211	0.02118	subject-2 enriched
s__	2.4642	0.03181	subject-2 enriched
__	1.9819	0.03474	subject-2 enriched
__	2.1263	0.04966	subject-2 enriched

Total significant genera: 15

Interpretation: Of the 15 taxa with q < 0.05, 12 are enriched in subject-2 (LFC from 1.78 to 5.41) and 3 are enriched in subject-1 (LFC -1.96, -3.53 and -4.29). The two most significant taxa (q = 2.3e-145 and 1.1e-124) have large positive LFC, and the strongest subject-1 taxon (LFC -4.29, q = 5.1e-56) is also highly significant, while the remaining taxa have q-values between 0.009 and 0.05. The genus column shows only "s__" because these taxa have no species-level name in SILVA, so the exact identity cannot be given from the output, and Figure 8 indicates that they belong mainly to Clostridia and Bacteroidia. In general, the two subjects have a clearly different gut microbial composition, with the subject-2 gut enriched in more taxa than the subject-1 gut.

Conclusion

The reads were almost exclusively bacterial. The rarefaction curves showed that the sequencing depth was sufficient for most samples. Alpha diversity was highest in the palms and lowest in the tongue (Shannon) and in the tongue and gut (Faith's PD), and the Bray-Curtis PERMANOVA (p = 0.001) together with the PCoA clusters showed that the body sites carry distinct communities. ANCOM-BC identified 15 taxa that differed significantly between the gut microbiota of the two subjects.